AX-200 Industrial Temperature Sensor

General description

The AX-200 is a digital temperature sensor with an I2C interface and a 12-bit resolution. It operates from 2.7 V to 5.5 V and draws less than 250 µA in active mode. An open-drain alert output signals when the temperature crosses a programmable limit.

Features

- Accuracy of ±0.5 °C from -10 °C to 85 °C
- Range of -55 °C to 150 °C
- Four selectable I2C addresses
- Standby current of 0.5 µA typical

Applications

- Industrial process control
- Battery chargers
- HVAC and building automation
- Server and rack monitoring

Electrical characteristics

Unless otherwise noted, limits apply at VDD = 3.3 V and an ambient temperature of 25 °C.

Parameter	Min	Typ	Max	Unit
Supply voltage	2.7	3.3	5.5	V
Supply current, active	–	180	250	µA
Supply current, standby	–	0.5	2	µA
Operating temperature	-55	–	150	°C
Accuracy, -10 °C to 85 °C	-0.5	±0.25	0.5	°C
Resolution	–	0.0078	–	°C
Conversion time	–	30	45	ms
I2C clock frequency	10	–	400	kHz

Pin description

Pin	Name	Function
1	VDD	Positive supply voltage
2	SDA	I2C data, open drain
3	SCL	I2C clock input
4	ALERT	Over-temperature alarm output, active low
5	GND	Ground

Programming example

The temperature register is read with a two-byte transfer. The following sketch assumes the default address of 0x48.

```
// Read the temperature register
Wire.beginTransmission(0x48);
Wire.write(0x00);
Wire.endTransmission();
Wire.requestFrom(0x48, 2);
int raw = (Wire.read() << 8) | Wire.read();
float celsius = raw / 128.0;
```

Ordering information

Part number	Package	Temperature grade	Packing
AX-200-DB	SOT-23, 5 pin	Industrial	Tape and reel, 3,000 pieces
AX-200-DM	MSOP, 8 pin	Industrial	Tube, 80 pieces
AX-200-AB	SOT-23, 5 pin	Automotive	Tape and reel, 3,000 pieces

Notes

1. All voltages are referenced to GND.
2. Accuracy excludes self-heating, which is below 0.1 °C in still air.
3. Do not exceed the absolute maximum ratings:
   - supply voltage 6.0 V;
   - voltage on any pin -0.3 V to VDD + 0.3 V.